

1 Customizing the Premium/Incentive Wage based on Formulas

1.1 Overview

You can use the incentive wage based on formulas to configure premium and incentive wage systems, which are based on the data recorded in HYDRA via the Order Data Collection ADE, the Time and Attendance PZE and the Incentive Wage LLE.

The customization can be performed by MPDV or by the customers that attended the relevant training.

The customization is made via user exits in HYDRA script language and user field configurations.

You always make the difference between an individual wage calculation and group incentives.

1.2 Requirements

1.2.1 Customization training

To customize a formula-based premium/incentive wage, you must attend the individual customizing training CUTI-LLE.

1.2.2 The HYDRA script language

You use the HYDRA script language to make customer-specific calculations or to assign values in user exits. You can also change data that has already been preprocessed in HYDRA.

A separate document describes the HYDRA script language in detail. In the sections below, knowledge of the script language is a precondition.

1.3 General data in user exits

1.3.1 Wage type data

In many user exits, master data for wage types is provided. They have the same name, but different prefixes (displayed below with an asterisk). It is always the same data that is provided for the wage types. The following sections therefore refer to this list:

Parameter	Type	Contents
Data for wage type		
*LART	C 4	Wage type

*BEZK	C 6	Short name
*BEZL	C 20	Detailed designation
*VAB	C 15	Responsibility area of wage type
*OPT_LLE	C 1	LLE indicator
*ZEIART	C 3	New: Time type
The following fields are in the first place relevant to the Time and Attendance, but the fields can also be evaluated in the Incentive Wage:		
*CERTIFY	C 1	Subject to approval J/N
*RM_LOBU	C 1	J/N: Confirm wage type to payroll system
*OPT_SZMA	C 1	Empty: not specified M: overtime S: target work I: undertime
*ART	C 1	G: basic wage Z: bonus
*PROZ	N	Percentage
*LART_LOBU	C 4	LOBU wage type. Wage type for the upload to payroll accounting
*MOD_LOBU	C10	LOBU indicator
*VERB	C 1	Processing
*LSS	F	Hourly rate
*OPT_KST	C 1	Selection indicator
*AVGART	C 1	Average Type
*RINT	N	Rounding interval in seconds
*RG	N	Rounding limit in seconds
*OPT_ADEABGL	C 1	J/N: Use wage type for ADE comparison
*OPT_ADEDEL	C 1	J/N: Delete PZE bookings to this wage type after ADE comparison

1.3.2 HR master data

The prefix of the HR master data is usually PNR_. An asterisk replaces this prefix in the table below:

Parameter	Type	Contents
*PNR	long	Personnel number
*DATB	date	Start of validity of the HR master data
*DATE	date	End of validity of the HR master data
*PNAME	char(40)	Last name
*PVORNAME	char(20)	First name
*NAME	char(62)	Last name, first name

*EINTRITT	date	Date of joining
*AUSTRITT	date	Date of leaving
*FIR	char(4)	Company
*BER	char(8)	Area
*KST	char(10)	Cost center.
*PKREIS	char(8)	Employee subgroup
*GEBDAT	date	Date of birth
*PRKZ	char(1)	Premium indicator
*ABT	char(8)	Department
*GESCHLECHT	char(1)	Gender M/W (male/female)
*INFOTEXT_01	char(40)	Free text field 01
...	...	...
*INFOTEXT_20	char(10)	Free text field 20
*INFOWERT_1	long	Free number field 1
...	...	...
*INFOWERT_5	long	Free number field 5
*INFODAT_1	date	Free date field 1
...	...	...
*INFODAT_5	date	Free date field 5
*LEISTGRP	char(8)	Regular premium group of person
*ANTFAKTLBON	long	Proport. factor for incentive bonus
*BPOS	char(10)	Regular operator function
*LPKZ	char(10)	Regular wage/premium indicator
*LART	char(4)	Regular wage type
*LGRP	char(4)	Regular wage group

1.3.3 Operation data

The prefix for operations is usually ANR_. An asterisk replaces this prefix in the table below:

Parameter	Type	Contents
*ANR	char(40)	Order number from posting
*AART	char(5)	Order type
*AARTKAT	char(2)	Category of order type
*PRKZ	char(1)	Piecework indicator
*LART	char(4)	Wage type
*TE	double	Single piece specification t_e

*TEB	double	Single piece specification (production resource) t_{eb}
*TR	double	Default setup time t_r
*TRB	double	Default setup time (production resources) t_{rb}
*SZY	double	Target cycle
*IMPFAKT	double	Pulse factor
*ATK	char(40)	Article
*MBVERH_NORM	double	M/O relation production (machine/operator relation)
*MBVERH_RUE	double	M/O relation setup (machine/operator rel.)
*KDAUNR	char(40)	Customer order number
*USERCODE	char(8)	User field key
*FU01	date	User field
...	...	...
*FU06	date	User field
*FU07	long	User field
...	...	...
*FU22	long	User field
*FU23	double	User field
...	...	...
*FU28	double	User field
*FU29	char(1)	User field
...	...	...
*FU44	char(1)	User field
*FU45	char(10)	User field
*FU46	char(10)	User field (former *PARAM_K1)
*FU47	char(10)	User field (former *PARAM_K2)
...	...	...
*FU50	char(10)	User field
*FU51	char(20)	User field
...	...	...
*FU53	char(20)	User field (former *BEM_1)
*FU54	char(20)	User field (former *BEM_2)
...	...	...
*FU57	char(20)	User field (former *KDPARAM_1)
*FU58	char(20)	User field (former *KDPARAM_2)
*FU59	char(20)	User field (former *KDPARAM_3)
*FU60	char(20)	User field (former *KDPARAM_4)

*FU61	char(20)	User field (former *KDPARAM_5)
...	...	...
*FU64	char(20)	User field
*FU65	char(40)	User field
*FU66	char(40)	User field
*SGR_GUTP	double	Target quantity primary quantity unit
*SGR_GUTS	double	Target quantity secondary quantity unit
*SGR_GUTT	double	Target quantity tertiary quantity unit
*SGR_GUTB	double	Target quantity base quantity unit
*SGR_AUSP	double	Planned scrap primary unit
*SGR_AUSS	double	Planned scrap primary quantity unit
*SGR_AUST	double	Planned scrap primary quantity unit
*SGR_AUSB	double	Planned scrap primary quantity unit

1.3.4 Machine/workplace data

The prefix for machines/workplaces is usually MNR_. An asterisk replaces this prefix in the table below:

Parameter	Type	Contents
*MNR	char(20)	Machine number
*PRKZ	char(1)	Premium indicator
*MGRP	char(20)	Machine group
*ART	char(1)	Type (single/group workplace)
*KST	char(10)	Regular cost center
*BEZK	char(8)	Short name
*BEZL	char(40)	Detailed designation
*BDEJMOD	long	Year model number
*IMPFAKT	long	Pulse factor
*FIR	char(4)	Company
*LEISTUNG	double	Planned performance level
*MSTDSATZ	double	Standard rate, machine
*PSTDSATZ	double	Standard labor rate
*TLG	long	Partitioning
*CAT	char(10)	Category
*VAB	char(15)	Responsibility area
*USERCODE	char(8)	User field key
*FU01	date	User field
...	...	...

*FU06	date	User field
*FU07	long	User field
...	...	...
*FU22	long	User field
*FU23	double	User field
...	...	...
*FU28	double	User field
*FU29	char(1)	User field
...	...	...
*FU44	char(1)	User field
*FU45	char(10)	User field
...	...	...
*FU50	char(10)	User field
*FU51	char(20)	User field
...	...	...
*FU64	char(20)	User field
*FU65	char(40)	User field
*FU66	char(40)	User field

1.3.5 Data of postings and bookings

This data is usually ADE log data. The prefix for this data is usually ADEPRO_. An asterisk replaces this prefix in the table below:

Parameter	Type	Contents
*VERWEIS	long	Database ID
*PNR	long	Personnel number
*KST	char(10)	Cost center
*DAT	date	Date
*DATB	date	Logon date
*ZEIB	long	Logon time
*DATE	date	Logoff date
*ZEIE	long	Logoff time
*LEISTGRP	char(8)	Premium group •
*LART	char(4)	Wage type
*MNR	char(20)	Machine
*ANR	char(40)	Order/operation
*TE	double	Single piece specification te
*TEB	double	Single piece specification (production resource) teb
*TR	double	Default setup time tr
*TRB	double	Default setup time (production resource) trb
*BEARB	char(10)	Modified by
*BEARBDAT	date	Modified on
*BEARBZEI	long	Processing time
*SART	char(10)	Record type of posting
*BPOS	char(10)	Operator position/function
*LPKZ	char(10)	Wage/premium indicator
*KARENZ	char(1)	Waiting period indicator (P=waiting period personnel, M=waiting period machine)
*SKNR	long	Shift number
*SCHICHT_DAT	date	Shift date
*BMK01	double	Order-related resource performance account 1
...	...	...
*BMK12	double	Order-related resource performance account 12
*DAUER	double	Order-related duration
*PBMK01	double	Person-related resource performance account 1
...	...	...

*PBMK12	double	Person-related resource performance account 12
*PDAUER	double	Labor time
*RGR_BMK01	double	Order-related remaining resource performance account 1
...	...	...
*RGR_BMK12	double	Order-related remaining resource performance account 12
*RGR_DAUER	double	Order-related remaining duration
*RGR_PBMK01	double	Person-related remaining resource performance account 1
...	...	...
*RGR_PBMK12	double	Person-related remaining resource performance account 12
*RGR_PDAUER	double	Remaining labor time
*GUT	double	Yield primary
*GUTP	double	Yield primary
*GUTS	double	Yield secondary
*GUTT	double	Yield tertiary
*GUTB	double	Yield base
*AUS	double	Scrap primary
*AUSP	double	Scrap primary
*AUSS	double	Scrap secondary
*AUST	double	Scrap tertiary
*AUSB	double	Scrap base
*NAC	double	Rework quantity primary (former *LEN)
*NACP	double	Rework quantity primary (former *LEN)
*NACS	double	Rework quantity secondary
*NACT	double	Rework quantity tertiary
*NACB	double	Rework quantity base
*PRB	double	Problem quantity primary
*PRBP	double	Problem quantity primary
*PRBS	double	Problem quantity secondary
*PRBT	double	Problem quantity tertiary
*PRBB	double	Problem quantity base
*EGG_GUT	long	Yield reason
*EGG_AUS	long	Scrap reason
*EGG_NAC	long	Rework reason
*EGG_PRB	long	Problem quantity reason
*EGR01	double	Recorded activity 1 to 10
*RGR01	double	Recorded remaining activity 1 to 10
*EGE01	char(3)	Activity unit 1 to 10

*VERWEIS_DLG_DATA	long	Database ID for dialog data
*CERTIFY	char(1)	Approval required
*SIGN	char(1)	Approved/rejected
*BEM	char(50)	Only with bonuses if data collection is activated (customization): comment
*USERCODE	char(8)	User field key
*FU01	date	User field
...	...	...
*FU06	date	User field
*FU07	long	User field
...	...	...
*FU22	long	User field
*FU23	double	User field
...	...	...
*FU28	double	User field
*FU29	char(1)	User field
...	...	...
*FU44	char(1)	User field
*FU45	char(10)	User field
...	...	...
*FU50	char(10)	User field
*FU51	char(20)	User field
...	...	...
*FU65	char(40)	User field
*FU66	char(40)	User field

1.3.6 Premium group data

Master data of the premium group usually has the prefix LEISTGRP_. An asterisk replaces this prefix in the table below:

Parameter	Type	Contents
*LEISTGRP	C 8	Premium group •
*BEZL	C 20	Premium group: name
*PRKZ	C 1	Premium group: premium indicator
*LART	C 4	Premium group: wage type (reserved)
LART_	*	Reserved: for master data of wage type, refer to the section

		further ahead. (Available as of November/2005)
*PRNR	N	Premium group: number of premium scheme
*WERT_01	F	Premium group: default value 1
...	...	...
*WERT_30	F	Premium group: default value 30
*PRMOD_01	C10	Premium group: mode 1
...	...	...
*PRMOD_05	C10	Premium group: mode 5
*VORGABE_01	F	Premium group: weekday-related default value 1 (for the relevant weekday)
...	...	...
*VORGABE_03	F	...
*VORGABE_04	C10	...

1.3.7 Time ticket data

The person day time tickets usually have the prefix L_. An asterisk replaces this prefix in the table below:

Parameter	Type	Contents
*PNR	N	Personnel number of time ticket
*ABREDAT	D	Settlement date of time ticket
*ANR	C 40	Operation number of time ticket
*ZEIART	C 3	Time type of time ticket. Only change in exceptional cases.
*LART	C 4	Wage type of time ticket
*TE	F	Target t_e of time ticket
*TEB	F	Target t_e of time ticket for production resource
*TR	F	Target t_r of time ticket
*TRB	F	Target t_r of time ticket for production resource
*DAUER	F	Duration of time ticket
*VGZ	F	Standard time of time ticket
*GUT	F	Yield of time ticket
*AUS	F	Scrap of time ticket
*NACHARB	F	Rework quantity of time ticket
*PROBLEM	F	Problem quantity of time ticket
*MENGE	F	Wage-relevant quantity of time ticket
*ZUSCHL	F	Bonus time of time ticket in seconds
*ZUSCHLGR	N	Bonus reason

*KST	C10	Cost center of time ticket
*LOHNGRP	C 4	Wage group of time ticket
*ZEITGRAD	F	Performance efficiency rate of time ticket in percent from standard processing. The performance efficiency rate does not integrate bonuses and deductions. Performance efficiency rates are only calculated for time tickets of the AKK time type (piecework). Changes of the performance efficiency rate are not transferred to the time ticket because this rate is always calculated using the standard and the actual time.
*REFNR	C 20	Reference number of time ticket from standard processing. Is usually empty, can be shown for customer-specific information in the time ticket log.
*LEISTGRP	C 8	Premium group •
*PRKTO_01 ... 30	F	Premium account of time ticket 1 to 30 that can be defined
*PRKZ_01 ... 05	C 20	Premium account of time ticket 1 to 5 that can be defined
*CERTIFY	C 1	Reserved: requires approval
*SIGN	C 1	Reserved: approved/rejected
*SOLLMENGE	F	Target quantity (yield / performance efficiency rate)
*SKNR	N	Shift number from ADE log record
*MNR	C 20	Machine
*BMKNR	N	Number of resource performance account for <i>time wage from production</i> .
*BEM	C 50	Comment
*DATB	D	Start date (from original record)
*ZEIB	N	Start time (from original record)
*DATE	D	End date (from original record)
*ZEIE	N	End time (from original record)

1.3.8 Data of person day performances

The person day performances usually have the prefix PNRTAG_. An asterisk replaces this prefix in the table below:

Parameter	Type	Contents
*DAT	D	Date
*PNR	N	Personnel number of time ticket
*FIR	C 4	Company
*ADE_DATB	D	Date of earliest ADE logon of person on this day
*ADE_ZEIB	N	Time of earliest ADE logon of person on this day in seconds
*ADE_DATE	D	Date of latest ADE logoff of person on this day
*ADE_ZEIE	N	Time of latest ADE logoff of person on this day in seconds

*VGZ	F	Piecework standard time of person on this day
*DAUER	F	Piecework actual time of person on this day
*LEISTGRAD	F	Piecework efficiency performance rate of person of this day
*LEISTGRAD_MIN	F	Smallest efficiency performance rate of time ticket of this day
*LEISTGRAD_MAX	F	Greatest efficiency performance rate of time ticket of this day
*ADE_DAUER	F	Sum total of the ADE time posted for this person
*PZE_DAUER	F	PZE attendance time of person (if PZE is used)
*LLE_DAUER	F	Sum total of time ticket duration of the Incentive Wage
*PRKTO_01 to *PRKTO_30	F	Premium accounts of day performance
*PRKZ_01 to *PRKZ_05	F	Premium indicator of day performance

1.4 Individual allocation

1.4.1 Overview

The individual allocation uses all ADE and LLE data that is **not** recorded for a premium group. If data is recorded with reference to a premium group, the group calculation is performed.

In general, the standard processing is performed before an individual allocation. With specific intermediate steps and at the end of the standard calculations, the values identified can be changed subsequently via user exit. The rules of the standard processing are described in the documents LLE-BP und LLE-ZGG.

Each relevant data record is separately calculated. The workflow of the individual allocation of time tickets is as follows:

1) Identifying the wage type

In standard processing, the wage type of the BDE posting is normally used. This wage type can be changed via user exit.

2) Identifying the time type

In the standard, the time type (AKK/ZL/...) is identified using the wage type, the order data, the HYDRA basic settings and the master data of machines and persons. The time type identified can subsequently be changed via user exit.

3) Standard processing

of BDE posting for time ticket. A direct intervention via user exit is not possible.

4) Recalculation of time tickets

The time tickets calculated by HYDRA in step 3 can be recalculated via user exit. For example, you can specify monetary evaluations in the available premium accounts that can be defined.

5) Identifying person day performance

6) Applying person day performance on time tickets

1.4.2 Combining PZE records

If the PZE time is used for the wage calculation and is not transferred from the daily PZE performance, then the PZE time is the result of several single records, e.g. PZE wage type postings. By default, HYDRA creates a PZE time ticket using each PZE original record. This can sometimes be inconvenient.

You can combine the PZE records via user exit that aggregates the data. The selected key information is then included in one single aggregated time ticket.

The user exit is called when the premium group resulting from *changes of group* has been entered in the PZE records.

The user exit always gets pairs of successive or parallel PZE records. The PZE records are processed for each person and settlement day one after the other using the start time.

PZE records that do not follow each other in time, are not combined to be processed as one.

You use the control variables of the user exit to specify the aggregation:

PZE1_SAVE	PZE2_SAVE	Action
1	1	Both PZE records are kept and updated. This is the default if no changes are made in the user exit.
1	0	PZE record 1 is updated, PZE record 2 is dropped
0	1	PZE record 2 is updated, PZE record 1 is dropped
0	0	Both PZE records are dropped and deleted.

User exit "hyl_pze_compr.hsc"

Import parameters:

Parameter	Type	Contents
PZE1_POS	N	Sequence number 1...n of first PZE record
PZE2_POS	N	Sequence number 1...n of second PZE record

Export parameters:

Parameter	Type	Contents
Data of first PZE record		
PZE1_SAVE	N	If set to 1, the first PZE record is saved.
PZE1_PNR	N	Personnel number of PZE record
PZE1_ABREDAT	D	Settlement date of PZE record
PZE1_LART	C 4	Wage type or payment day type of PZE record
PZE1_KST	C10	Cost center of PZE record
PZE1_LEISTGRP	C 8	Premium group of PZE record
PZE1_DATB	D	Start date (from original record)
PZE1_ZEIB	N	Start time (from original record)
PZE1_DATE	D	End date (from original record)
PZE1_ZEIE	N	End time (from original record)
PZE1_DAUER	F	Duration of PZE record in hours or seconds (system setting with LLE 7.2)
Data of second PZE record		
PZE2_SAVE	N	If set to 1, the second PZE record is saved.
PZE2_PNR	N	Personnel number of PZE record
PZE2_ABREDAT	D	Settlement date of PZE record
PZE2_LART	C 4	Wage type or payment day type of PZE record
PZE2_KST	C10	Cost center of PZE record
PZE2_LEISTGRP	C 8	Premium group of PZE record
PZE2_DATB	D	Start date (from original record)
PZE2_ZEIB	N	Start time (from original record)
PZE2_DATE	D	End date (from original record)
PZE2_ZEIE	N	End time (from original record)
PZE2_DAUER	F	Duration of PZE record in hours or seconds (system setting with LLE 7.2)

Example:

```
hydra basic;

/* -----
Script  : hyl_pze_compr.hsc
Descr.: Aggregation of PZE records

The user exit always gets pairs of successive
or parallel PZE records. The PZE records are processed for each person and
settlement day one after the other using the start time.

PZE records that do not follow each other in time, are not
combined to be processed as one.
```

You use the control variables to specify the aggregation:

```

PZE1_SAVE PZE2_SAVE action
-----
1          1      Both PZE records are kept and updated.
1          0      PZE record 1 is updated, PZE record 2 is dropped
0          1      PZE record 2 is updated, PZE record 1 is dropped
0          0      Both PZE records are dropped and deleted.

$Revision: 1.0 $
$Date: 2006/12/14 18:03:26 $
----- */

// control information
export PZE1_SAVE      long      ; // If set to 1, the first PZE record is saved.
import PZE1_POS       long      ; // Sequence number 1...n of first PZE record
export PZE2_SAVE      long      ; // If set to 1, the second PZE record is saved.
import PZE2_POS       long      ; // Sequence number 1...n of second PZE record

// Variables of first PZE record
export PZE1_PNR       long      ; // Personnel number of PZE record
export PZE1_ABREDAT   date      ; // Settlement date of PZE record
export PZE1_LART      char(4)   ; // Wage type or payment day type of PZE record
export PZE1_KST       char(10)  ; // Cost center of PZE record
export PZE1_LEISTGRP  char(8)   ; // Premium group of PZE record
export PZE1_DATB      date      ; // Start date (from original record)
export PZE1_ZEIB      long      ; // Start time (from original record)
export PZE1_DATE      date      ; // End date (from original record)
export PZE1_ZEIE      long      ; // End time (from original record)
export PZE1_DAUER     double    ; // Duration of PZE record in hours or seconds
                                ; // (System settings with LLE 7.2)

// Variables of second PZE record
export PZE2_PNR       long      ; // Personnel number of PZE record
export PZE2_ABREDAT   date      ; // Settlement date of PZE record
export PZE2_LART      char(4)   ; // Wage type or payment day type of PZE record
export PZE2_KST       char(10)  ; // Cost center of PZE record
export PZE2_LEISTGRP  char(8)   ; // Premium group of PZE record
export PZE2_DATB      date      ; // Start date (from original record)
export PZE2_ZEIB      long      ; // Start time (from original record)
export PZE2_DATE      date      ; // End date (from original record)
export PZE2_ZEIE      long      ; // End time (from original record)
export PZE2_DAUER     double    ; // Duration of PZE record in hours or seconds
                                ; // (System settings with LLE 7.2)

//-----
long main()
{
    dprint( "comparison"||(PZE1_POS using "##&")||" "||(PZE1_DATB||" "||(PZE1_ZEIB using "$TIME")||
        " - "||(PZE1_DATE||" "||(PZE1_ZEIE using "$TIME")||", "||(PZE1_DAUER using "#&.&")||" h" );
    dprint( "      with"||(PZE2_POS using "##&")||" "||(PZE2_DATB||" "||(PZE2_ZEIB using "$TIME")||
        " - "||(PZE2_DATE||" "||(PZE2_ZEIE using "$TIME")||", "||(PZE2_DAUER using "#&.&")||" h" );

    if( ( PZE1_PNR      = PZE2_PNR      ) and
        ( PZE1_ABREDAT  = PZE2_ABREDAT  ) and
        ( PZE1_LART     = PZE2_LART     ) and
        ( PZE1_KST      = PZE2_KST      ) and
        ( PZE1_LEISTGRP = PZE2_LEISTGRP ) )
    {
        PZE1_SAVE = 1;
        PZE2_SAVE = 0;
        PZE1_DATE = PZE2_DATE;
        PZE1_ZEIE = PZE2_ZEIE;
        PZE1_DAUER = PZE1_DAUER + PZE2_DAUER;
        dprint( "      new"||(PZE1_POS using "##&")||" "||(PZE1_DATB||" "||(PZE1_ZEIB using "$TIME")||
            " - "||(PZE1_DATE||" "||(PZE1_ZEIE using "$TIME")||", "||(PZE1_DAUER using "#&.&")||" h" );
    }

    return 0;
}
//-----

```

1.4.3 Identifying the wage type

In standard processing, the wage type of a time ticket is transferred from the original BDE personnel posting (B record).

This original BDE personnel posting is identified using the wage type stored for the operation. Use the following user exit to change the wage type that is identified via standard processing.

Note that the identified wage type specifies if piecework or time wage applies. Also refer to the LLE basic settings and explanations in the document LLE-BP and the sections below.

The user exit is only called when time tickets are created via BDE personnel postings and when bonuses/deductions are created. With other original records, you can only change the wage type using the user exit lsv00000.hsc.

User exit "lsv00000.hsc"

Import parameters:

Parameter	Type	Contents
ART	C10	Source of time ticket PB : time ticket from ADE personnel postings PZ : time ticket from PZE wage type posting ZUS : time ticket from bonus
PNR	N	Person: personnel number
PNR_*	*	Data of person (see general description above)
ANR	C 40	Order number from personnel posting
ANR_*	*	Operation data (see general description above)
ANR_LART_*	*	Master data of wage type included in operation (see general description above)
MNR	C 20	Machine number from personnel posting
MNR_*	*	Master data of machine (see general description above)
ADEPRO_*	*	Data of posting (see general description above)
ADEPRO_LART_*	*	Master data of wage type included in posting (see general description above)
LEISTGRP_*	*	Master data of premium group (see general description above) Available as of hyl_compute.exe out 8.1.1.93 (04/2018)

Export parameters:

Parameter	Type	Contents
LART	C 4	Wage type of time ticket, the wage type of the personnel posting is prepopulated (standard processing)

1.4.4 Identifying the time type

The user exit is only called when time tickets are created via BDE personnel postings and when bonuses/deductions are created. With other original records, you can only change the time type using the user exit lsv00000.hsc.

The time type specifies how the time ticket is calculated. Only time tickets with time type AKK (piecework) are calculated as piecework and are assigned a performance efficiency rate by HYDRA.

The following time types are available (see also documentation LLE-BP):

Time type	Meaning
AKK	Piecework Only piecework time tickets get a performance efficiency rate
ZUS	Bonuses and deductions Time tickets of bonuses and deductions recorded
ZL	Time wage Time tickets with time wage can be generated for production orders that are not calculated using a piecework wage type or for production orders with piecework that had malfunction times.
EA	On-the-job training Rarely used time type. Is created if persons of the HR master data have the indicator "On-the-job training".
GK	Overheads This time type results from the editing of overhead orders.
KAR	Waiting period Results from postings generated via the HYDRA waiting period processing.
GRP	Group bonus Results from times posted for premium groups that are also stored for the individual allocation.
PZE	Labor time from PZE Results from times that were recorded and calculated via the HYDRA Time & Attendance PZE.

User exit "lsz00000.hsc"

Import parameters:

Parameter	Type	Contents
ART	C10	Source of time ticket PB : time ticket from ADE personnel postings PZ : time ticket from PZE wage type posting ZUS : time ticket from bonus
PNR	N	Person: personnel number
PNR_*	*	Data of person (see general description above)
ANR	C 40	Order number from personnel posting
ANR_*	*	Operation data (see general description above)
ANR_LART_*	*	Master data of wage type included in operation (see general description above)
MNR	C 20	Machine number from personnel posting
MNR_*	*	Master data of machine (see general description above)
ADEPRO_*	*	Data of posting (see general description above)
ADEPRO_LART_*	*	Master data of wage type included in posting (see general description above)
LART	C 4	Wage type of time ticket, prepopulated using the wage type of the time ticket
TLS_LART*	*	Master data of wage type included in time ticket (see general description above)
LEISTGRP_*	*	Master data of premium group (see general description above) Available as of hyl_compute.exe out 8.1.1.93 (04/2018)

Export parameters:

Parameter	Type	Contents
ZEIART	C 3	Time type of time ticket, prepopulated using standard processing.

1.4.5 Recalculation of time tickets

When HYDRA has performed all steps of the standard processing of a time ticket, you can use the following user exit to recalculate the time ticket.

User exit "lsv00000.hsc"

Import parameters:

Parameter	Type	Contents
ART	C10	Source of time ticket PB : time ticket from ADE personnel postings PZ : time ticket from PZE wage type posting ZUS : time ticket from bonus
PNR	N	Person: personnel number
PNR_*	*	Data of person (see general description above)
ANR	C 40	Order number from personnel posting
ANR_*	*	Operation data (see general description above)
MNR	C 20	Machine number from personnel posting
MNR_*	*	Master data of machine (see general description above)
ADEPRO_*	*	Data of posting (see general description above)
LART*	*	Master data of wage type included in time ticket (see general description above)
LEISTGRP_*	*	Master data of premium group (see general description above) Available as of hyl_compute.exe out 8.1.1.93 (04/2018)

Export parameters:

All export parameters are prepopulated using the results of the standard processing. In particular the premium accounts and premium indicators can be used to make separate calculations. Only change the other export parameters in exceptional cases via user exit.

Parameter	Type	Contents
L_*	*	Data for time ticket, see section above.

1.4.6 Importing time tickets into person day performance

The time tickets calculated using the available data, are imported into the person day performance:

User exit "hyl_tls2pnrtag.hsc"

Import parameters:

Parameter	Type	Contents
PNR	N	Person: personnel number
PNR_*	*	Data of person, see section above.
ANR	C 40	Operation number of time ticket
ANR_*	*	Operation data, see section above.
MNR	C 20	Machine of time ticket
MNR_*	*	Machine data, see section above.
TLS_*	*	Data of time ticket, see section above.
TLS_LART_*	*	Wage type data of time ticket, see section above

Export parameters:

Parameter	Type	Contents
PNRTAG_*	*	Data of person day performance
BUFFER_1 and BUFFER_2	C 32000	Free buffer variables. The content of these variables is kept during the complete calculation of the person day performance. Using these variables, you can save values in BAPI format from the import of time tickets into the day performance up to the distribution of the day performance on time tickets. When the calculation of a person day performance is started, the buffer variables are emptied.

This user exit requests two functions:

Function	Task
main()	Distributes the data of a time ticket into the fields of the person day performance
final_calc()	Final calculation of the person day performance. In this case, only the import/export variables of the person day performance are useful. The time ticket data remain empty.

1.4.7 Using person day performance with time tickets

The person day performances can then be used for the time tickets to make calculations that require day totals.

One example is the calculation of a performance efficiency rate if specific proportions of the time wage must be deducted from the total labor time of the day to calculate the actual duration. The day-related performance efficiency rate can then be assigned to the separate piecework time tickets.

If required, you can also use this user exit to change the person day performance.

User exit "hyl_pnrtag2tls.hsc"

Import parameters:

Parameter	Type	Contents
PNR	N	Person: personnel number
PNR_*	*	Data of person, see section above.
ANR	C 40	Operation number of time ticket
ANR_*	*	Operation data, see section above.
MNR	C 20	Machine of time ticket
MNR_*	*	Machine data, see section above.
TLS_LART_*	*	Wage type data of time ticket, see section above

Export parameters:

Parameter	Type	Contents
TLS_*	*	Data of time ticket, see section above.
PNRTAG_*	*	Data of person day performance
BUFFER_1 and BUFFER_2	C 32000	Free buffer variables. The content of these variables is kept during the complete calculation of the person day performance. Using these variables, you can save values in BAPI format from the import of time tickets into the day performance up to the distribution of the day performance on time tickets. When the calculation of a person day performance is started, the buffer variables are emptied.

1.5 Group allocation

1.5.1 Overview

The postings for orders and persons recorded via ADE are the data basis of the premium wage based on formulas. These postings include the run times, separated into main production time and malfunction times. And the quantities produced. These postings also include the wage specifications like wage type, t_e , t_{eb} and t_r . Here, they are checked and can be manually corrected and changed for each separate posting, if required. Bonuses and deductions are also integrated in the calculation.

The data is recorded for machines, orders and persons.

The illustration below helps to understand the interrelations. The illustration shows a sample premium system.

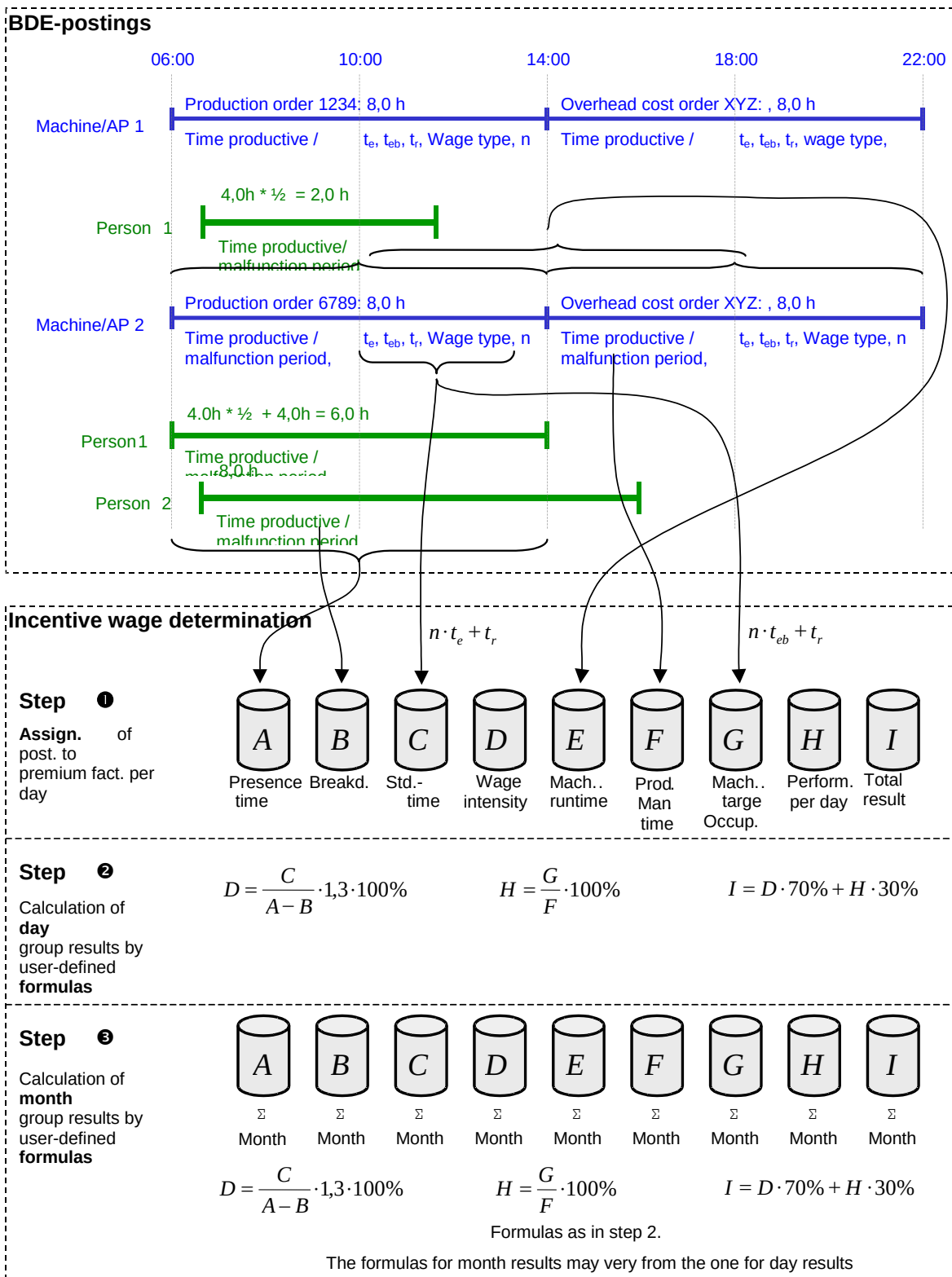


Illustration: Schema to illustrate the premium wage calculation based on formulas in HYDRA.

Explanatory notes:

HYDRA provides premium accounts that you are free to define for the calculation of premium wages based on formulas. The accounts have the letters A to I in the illustration. You use these accounts to record data like standard times or actual times. And you can calculate these premium accounts using other premium accounts.

It is possible to store different schemes for different forms of premiums.

In general, you calculate premium wages in three steps.

1. Sorting of the posted data into premium accounts on a daily basis

In the first step, you control the sorting of the data from postings into the premium accounts via script language. In the illustration, this is shown with the arrows. The arrows only represent one posting in the illustration. But of course, all postings of the same type are sorted the same way:

- Premium account **A**: the account records the total personnel processing time that results from the personnel postings.
- Premium account **B**: this account records for each premium group the total *malfunction time* that was included in the personnel processing time resulting from personnel postings.
- Premium account **C**: this account records for each premium group the person-related standard time for the personnel postings. The standard time is calculated using the data of the postings via the formula $n * t_e + t_r$.
- Premium account **E**: this account records for each premium group the total order run time at the machines.
- Premium account **F**: this account records for each premium group the productive order run time at the machines (based on production orders, without malfunction times).
- Premium account **G**: this account records for each premium group the order/machine-related standard time using the order postings. The standard time is calculated using the data of the postings via the formula $n * t_{eb} + t_r$.

2. Calculation of daily group results

In this step, you can calculate daily interim results using the data recorded in the premium accounts when all postings of a premium group have been processed for the day. The illustration shows the formulas used in the example. The premium accounts **D**, **H** and **I** are calculated using the formulas. The user can define the formulas via script language.

3. Calculation of monthly group results

In this step, you can make calculations based on the premium accounts that are totaled on a monthly basis. To this end, the same formulas are used as in step 2. The results are then assigned again to the dependent premium accounts **D**, **H** and **I**. Also other formulas or extended calculations can be implemented. The user can define the formulas via script language.

The flexibility of the script language allows not only simple calculations of a performance efficiency rate, but you can also realize monetary evaluations. As part of an implementation support, you can map existing premium models.

In LLE version 7.2, another step is available. You can also assign the result calculated for the group on the day time tickets of the persons belonging to the relevant group. This assignment is only made if the relevant user exit is available.

1.5.2 Step 1: distribution of data to premium accounts

User exit "lpv00000.hsc"

1.5.2.1 Initialization

User exit "lpv1000.hsc", function init().

At start of processing, the function **init()** is called in user exit lpv1000.hsc. In this function, you use the export variables **USES_ART** to control which data types are processed in the calculation. Only the data required to calculate the wage data must be requested.

The import and export variables are described in the section below.

With the function **init()** it is important to make sure that the import variables are not filled. The import variables are empty or 0.

```
...
export USES_ART char(50);
...

/*-----*/
long init()
{
    // The export variable USES_ART controls which data is processed.
    // USES_ART = "AU AE PB ZUS TLS MDE";
    // AU : U records (ADE order interruption)
    // AE : E records (ADE order end)
    // PB : B records (ADE personnel postings)
    // ZUS : Bonuses (from LLE)
    // TLS : Time tickets (also include PZE times from wage types for group incentives)
    // [MDE : MDE log data, not available in the standard]
    // You cannot explicitly control the group result that is used to calculate bonuses (type "GRP").
    USES_ART = "ZUS TLS"; // Only time tickets and bonuses are required.

    return 0;
}
```

1.5.2.2 Data distribution to premium accounts

User exit "lpv00000.hsc"

Import parameters:

Note: The column KAR specifies the data that results from BDE waiting period processing with order and personnel postings. You can identify these postings via the order type ANR_AART.

Parameter	Type	Contents	AU/ AE/ PB	KAR	ZUS
ART	C 4	Posting type. "AU"/"AE": order posting interruption/end "PB": personnel posting (operator) "ZUS": bonus "GRP": group result to calculate incentives. Here, no data of orders, machines, persons, bonus reasons and postings/bookings is available. The variable PNR_LEISTGRP includes the ID	X	X	X

		of the premium area. The variables of premium group LEISTGRP* include the master data of the premium group (not of the premium area). The export parameters are prepopulated using the results of the premium groups . They can be changed in the script. After execution of the script, the export parameters are added to the area result . "TLS": time ticket with premium group. With time ticket, special assignment of the fields ADEPRO_*, see below			
Order data					
ANR	C 40	Order number	X	X	X
ANR_AART	C 5	Operation: order type: 0 = PPS order, production order 1 = overhead order 2 = rework order 3 = capacity OP 4 = overhead order type II ... other customer-specific order types	X	X	X
ANR_AARTKAT	C 2	Operation: category of order type	X	X	X
ANR_SZY	N	Operation: target cycle of machine in seconds per 1000 cycles.	X		X
ANR_TLG	N	Operation: target partitioning of machine	X		X
ANR_DATB	D	Operation: date of first logon	X		X
ANR_ZEIB	N	Operation: time of first logon in seconds	X		X
ANR_PRKZ	C 1	Operation: premium indicator	X		X
ANR_PARAM_K1	C 8	Operation: free parameter, text 1	X		X
ANR_PARAM_K2	C 8	Operation: free parameter, text 2	X		X
ANR_KDPARAM_1	C 100	Operation: general information	X		X
ANR_KDPARAM_2	C 20	Operation: general information 2	X		X
...	...	...			
ANR_KDPARAM_5	C 20	Operation: general information 5	X		X
ANR_LART	C 4	Operation: planned wage type	X		X
ANR_LART_*	*	Master data of wage type, refer to section further ahead.	X		X
ANR_TE	F	Operation: planned standard time t_e in seconds per 1000 pieces	X		X
ANR_TR	F	Operation: planned setup specification t_r in seconds	X		X
ANR_TEB	F	Operation: planned standard time t_{eb} in seconds per 1000 pieces	X		X
ANR_ADATF	D	Operation: planned earliest start date	X		X

ANR_AZEIF	N	Operation: planned earliest start time in seconds	X		X
ANR_ADATB	D	Operation: planned start date	X		X
ANR_AZEIB	N	Operation: planned start time in seconds	X		X
ANR_ADATB	D	Operation: planned end date	X		X
ANR_AZEIE	N	Operation: planned end time in seconds	X		X
ANR_ADATS	D	Operation: planned latest end date	X		X
ANR_AZEIS	N	Operation: planned latest end time in seconds	X		X
ANR_ATK	C 25	Operation: article number	X		X
ANR_PARAM_1	N	from 02/06: free parameter, value 1	X		X
ANR_PARAM_2	N	from 02/06: free parameter, value 2	X		X
ANR_PARAM_3	N	from 02/06: free parameter, value 3	X		X
ANR_BEM_1	C 15	from 02/06: comment 1	X		X
ANR_BEM_2	C 15	from 02/06: comment 2	X		X
ANR_KDBEZ	C 16	from 02/06: customer name	X		X
ANR_MBVERH_NORM	F	from 02/06: number of employees production	X		X
ANR_MBVERH_RUE	F	from 02/06: number of employees setup	X		X
ANR_OPTKRIT	C 20	from 02/06: optimization criteria	X		X
ANR_OPTKZ	C 1	Operation: optimization identicator	X		X
ANR_COLOR	C 20	from 02/06: color	X		X
ANR_FU01	D	from 02/06: user field in ADE 7.2	X		X
...					
ANR_FU06	D	from 02/06: user field in ADE 7.2	X		X
ANR_FU07	N	from 02/06: user field in ADE 7.2	X		X
...					
ANR_FU22	N	from 02/06: user field in ADE 7.2	X		X
ANR_FU23	F	from 02/06: user field in ADE 7.2	X		X
...					
ANR_FU28	F	from 02/06: user field in ADE 7.2	X		X
ANR_FU29	C 1	from 02/06: user field in ADE 7.2	X		X
...					
ANR_FU44	C 1	from 02/06: user field in ADE 7.2	X		X
ANR_FU45	C10	from 02/06: user field in ADE 7.2	X		X
...					
ANR_FU50	C10	from 02/06: user field in ADE 7.2	X		X
ANR_FU51	C 20	from 02/06: user field in ADE 7.2	X		X
...					

ANR_FU63	C 20	from 02/06: user field in ADE 7.2	X		X
ANR_FU65	C 40	from 02/06: user field in ADE 7.2	X		X
ANR_FU66	C 40	from 02/06: user field in ADE 7.2	X		X

Data of person					
PNR	N	Person: personnel number	X	X	X
PNR_PRKZ	C 1	Person: premium indicator	X	X	X
PNR_ABT	C 8	Person: department	X	X	X
PNR_BER	C 8	Person: area	X	X	X
PNR_EINTRITT	D	Person: date of joining	X	X	X
PNR_FIR	C 4	Person: company	X	X	X
PNR_KST	C10	Person: cost center	X	X	X
PNR_LEISTGRP	C 8	Person: regular premium group of person	X	X	X
PNR_ANTFAKTLBON	N	Person: proport. factor for incentive bonus	X	X	X
PNR_GEBDAT	D	Person: date of birth	X	X	X
PNR_GESCHLECHT	C 1	Person: gender M/W (male/female)	X	X	X
PNR_INFODAT_1	D	Person: free date field 1	X	X	X
...	...	...			
PNR_INFODAT_5	D	Person: free date field 5	X	X	X
PNR_INFOTXT_01	C 40	Person: free text field 1	X	X	X
...	...	...			
PNR_INFOTXT_20	C 40	Person: free text field 20	X	X	X
PNR_INFOWERT_1	N	Person: free number field 1	X	X	X
...	...	...			
PNR_INFOWERT_5	N	Person: free number field 5	X	X	X
Machine data					
MNR	C 20	Machine number	X	X	X
MNR_PRKZ	C 1	Machine: premium indicator	X	X	X
MNR_LEISTGRP	C10	Machine: premium group populated by default	X	X	X
MNR_*	*	Master data of machine (see general description above) Important: this general master data is available only as of program versions 06/2010. In older versions, only the three fields mentioned above are available as machine master data.			

Data of premium group Also with type = "GRP" (distribution of premium group results to the accounts of the premium areas) the master data of the premium group is included here.					
LEISTGRP	C10	Premium group •	X	X	X
LEISTGRP_*	*	Data of premium group, see section above.	X	X	X
Data of bonuses and bonus reasons					
CERTIFY	C 1	Requires approval J/N If the bonus actually requires approval, but the option <i>Allocate if still subject to authorization</i> is activated, then the bonus does not require approval here and the parameter CERTIFY is set to "N".			X
SIGN	C 1	Approved J/N/A A=rejected			X
ZUSCHLGR	N	Bonus reason			X
ZUSCHLGR_SZ	C 1	Bonus reason: is bonus for target or actual time? (S/I)			X

Date from posting/booking Note: special assignment with TYPE="TLS", see separate table below.					
ADEPRO_DAT	D	Date	X	X	X
ADEPRO_BMK01	F	Resource performance account 1 in hours	X	X	
...			X	X	
ADEPRO_BMK11	F	Resource performance account 11 in hours. With bonuses, here bonus duration.	X	X	X
ADEPRO_BMK12	F	Resource performance account 12 in hours	X	X	
ADEPRO_PDAUER	F	Labor time in hours. With bonuses, bonus duration.	X	X	X
ADEPRO_KST	C10	Cost center	X	X	
ADEPRO_TE	F	t _e recorded in hours per 1000 pieces	X		
ADEPRO_TR	F	t _r recorded in hours	X		
ADEPRO_TEB	F	t _{eb} recorded in hours per 1000 pieces	X		
ADEPRO_LART	C 4	Wage type	X		
ADEPRO_LART_*	*	Master data of wage type, refer to section further ahead. (Available as of November/2005)	X		
ADEPRO_GUT	F	Yield (primary quantity)	X		
ADEPRO_AUS	F	Scrap quantity (primary quantity)	X		
ADEPRO_NAC	F	Rework quantity (primary quantity)	X		
ADEPRO_PRB	F	Problem quantity (primary quantity)	X		
ADEPRO_DATB	D	Logon date	X		X

ADEPRO_ZEIB	F	Logon time in seconds	X		X
ADEPRO_DATE	D	Logoff date	X		X
ADEPRO_ZEIE	F	Logoff time in seconds	X		X
ADEPRO_SKNR	N	Shift number	X	X	
ADEPRO_BPOS	C10	Operator position/function	X		
ADEPRO_LPKZ	C10	Premium indicator (wage group)	X		
ADEPRO_USER_01 to ADEPRO_USER_05	C 40	Customer-specific log data. See notes below. Note: in the script, the function init() controls the assignment of these variables; for this reason, define these variables as export variables.			

Export parameters:

Parameter	Type	Contents
DAUER	F	Duration in hours
VORG	F	Standard time in hours
ZUSCHL	F	Bonuses in hours. Normally used for bonuses that refer to the standard time.
UPZ	F	Non-productive time in hours
AUSFZ	F	Downtime in hours
GKZ	F	Overhead cost times in hours
GUTSCHR	F	Bonus time in hours, normally used for bonuses that refer to the actual time.
WARTEN	F	Waiting time in hours
PRKTO01	F	Value that is allocated to premium account 1
...	...	...
PRKTO30	F	Value that is allocated to premium account 30
PRKZ01	C 20	Premium indicator 1. Prepopulated using the previously set premium indicator
...	...	...
PRKZ05	C 20	Premium indicator 5, see above

Special assignment of data from posting/booking in the processing of time tickets.

In the processing of group time tickets, the time tickets are passed to the user exit in the structure of the ADE postings in order to limit the number of import/export variables. In this case, a special assignment of the fields ADEPRO_* is used.

Data from posting/booking					
ADEPRO_DAT	D	Date	x		
ADEPRO_BMK01	F	Value of premium account 1 of time ticket. Because of conversion rules of the past, this value must be multiplied by 0,0036 for further use ($3600 / 1000000 = 0,0036$).	x		
...			x		
ADEPRO_BMK10	F	Value of premium account 10 of time ticket. Because of conversion rules of the past, this value must be multiplied by 0,0036 for further use.	x		
ADEPRO_BMK11	F	Standard time of time ticket	x		
ADEPRO_BMK12	F	Fixed 0.	-		
ADEPRO_PDAUER	F	Duration of time ticket	x		
ADEPRO_KST	C10	Cost center	x		
ADEPRO_TE	F	t_e in hours per 1000 pieces	x		
ADEPRO_TR	F	t_r in hours	x		
ADEPRO_TEB	F	Always 0.	-		
ADEPRO_LART	C 4	Wage type	x		
ADEPRO_LART_*	*	Master data of wage type, refer to section further ahead.	x		
ADEPRO_GUT	F	Yield	x		
ADEPRO_AUS	F	Scrap	x		
ADEPRO_NAC	F	Always 0	-		
ADEPRO_PRB	F	Always 0	-		
ADEPRO_DATB	D	Logon date	x		
ADEPRO_ZEIB	F	Logon time in seconds	x		
ADEPRO_DATE	D	Logoff date	x		
ADEPRO_ZEIE	F	Logoff time in seconds	x		
ADEPRO_SKNR	N	Shift number	x		
ADEPRO_BPOS	C10	Always empty.	-		
ADEPRO_LPKZ	C10	Time type of time ticket	x		
ADEPRO_USER_01 to ADEPRO_USER_05	C 40	Always empty.	-		
Further data					
ZUSCHLGR	N	Bonus reason	x		

1.5.2.3 User-specific log data

Using the export variables ADEPRO_USER_01 to ADEPRO_USER_05, you can additionally select customer-specific data from the database (ADE log data, order backlog data or other master data). This can be user fields of the log data or of the machines, for example. You require detailed knowledge of the database structures and the internal processes of wage calculation to this end.

You initialize the fields using the user exit described above "lvp00000.hsc" by calling the function **init()**. Assign SQL fragments (column names) to the variables of data that must be selected additionally. When the function **main()** is called in the following requests, the variables then include the relevant data. In the function **init()**, SQL fragments are assigned to the variables ADEPRO_USER_01 to ADEPRO_USER_05. For this reason, they must be declared as export variables.

Available database tables:

Table	Alias	Contents
ade_protokoll	ap.	ADE log data
auftrag_status	ast.	Status information on the operation
maschinen	m.	Maschine master data
lle_leist_grp	lg.	Master data of premium groups

If you select columns that do not have data type "char(n)", you must phrase the column so that the database selects this column as char(n). With Oracle and SQL server, you can use the relevant type conversions according to the database used. See example below. If you do not respect this, a database error is produced because of a UNION select.

Example:

```
...
export ADEPRO_USER_01      char(40);
export ADEPRO_USER_02      char(40);
export ADEPRO_USER_03      char(40);
export ADEPRO_USER_04      char(40);
export ADEPRO_USER_05      char(40);
...

/*-----*/
/* Init                                         */
/*-----*/
long init()
{
    ADEPRO_USER_01 = "ap.user_c_29"; // Transaktionscode
    ADEPRO_USER_02 = "ap.user_c_45"; // Buchungsschlüssel
    // MS-SQL-Server:
    ADEPRO_USER_03 = "CAST( ap.user_n_17 AS varchar(40)), "; // bonus points
    // Oracle:
    // ADEPRO_USER_03 = "TO_CHAR( ap.user_n_17 )"; // bonus points
    dprint( "Init ADEPRO-Userfelder Transaktionscode und Buchungsschlüssel" );
}
```

```

    return 0;
}

/*-----*/
/* main function                                     */
/*-----*/
long main()
{
    ...
    if( ADEPRO_USER_01 = "S" ) // transaction code
    {
        if( ADEPRO_USER_02 = "JFLF" ) // transaction key
        {
            PRKTO14 = LEISTGRP_WERT_01 / 60; // Ruesten DI [default in min]
            PRKTO15 = char2long( ADEPRO_USER_03 ); // bonus points
        }
    }
    ...
}

```

1.5.3 Step 2+3: daily/monthly calculation of premium accounts

User exit "lpb00000.hsc"

Import parameters:

Note: the person-related import parameters are only filled for the person (PNR_xxx), if the user exit is called for the list *Personal group participation*.

Parameter	Type	Contents
ART	C 1	T: script is run for daily calculation M: script is run for monthly calculation P: script is run to calculate the personal group participation.
DAT	D	Date. With type P or M, the date is the last day of the period used for the calculation (end of month).
MDEFEITG	C 1	With type T: Is the day stored as public holiday in the MDE public holidays J/N Otherwise: empty
PDAUER	N	With type P: working time of a person in the premium group in seconds. Otherwise: 0.
Data of premium group When premium areas are calculated, the premium area master data is included.		
LEISTGRP	C10	Premium group •
LEISTGRP_*	*	Data of premium group, see section above.
Data of person They are only filled if the user exit is called for the list <i>Personal group participation</i> .		
PNR	N	Person: personnel number
PNR_PRKZ	C 1	Person: premium indicator
PNR_ABT	C 8	Person: department

PNR_BER	C 8	Person: area
PNR_EINTRITT	D	Person: date of joining
PNR_FIR	C 4	Person: company
PNR_GEBDAT	D	Person: date of birth
PNR_GESCHLECHT	C 1	Person: gender M/W (male/female)
PNR_INFODAT_1	D	Person: free date field 1
...	...	...
PNR_INFODAT_5	D	Person: free date field 5
PNR_INFOTXT_01	C 40	Person: free text field 1
...	...	...
PNR_INFOTXT_20	C 40	Person: free text field 20
PNR_INFOWERT_1	N	Person: free number field 1
...	...	...
PNR_INFOWERT_5	N	Person: free number field 5
PNR_KST	C10	Person: cost center
PNR_LEISTGRP	C 8	Person: regular premium group of person
PNR_ANTFAKTLBON	N	Person: proport. factor for incentive bonus

Export parameter:

Parameter	Type	Contents
VORG	N	Standard time in seconds
ZUSCHL	N	Bonuses in seconds. Is normally used for bonuses that refer to the standard time.
UPZ	N	Non-productive time in seconds
DAUER	N	Duration in seconds
AUSFZ	N	Downtime in seconds.
GKZ	N	Overhead cost times in seconds
GUTSCHR	N	Time credit in seconds. Is normally used for bonuses that refer to the actual time.
WARTEN	N	Waiting time in seconds
LEISTGRAD	F	Performance level in percent
PRKTO01	F	Premium account 1
...	...	...
PRKTO20	F	Premium account 20
PRKZ01	C 20	Premium indicator 1
...	...	...

PRKZ05	C 20	Premium indicator 5
--------	------	---------------------

1.5.4 Group time tickets

For the labor time, group time tickets are created. You can change these time tickets via the user exit "lsv00000.hsc": "Recalculation of time tickets". This user exit is described in a paragraph above in section "Individual allocation". This user exit used the time type "GRP" for group time tickets.

1.5.5 Assigning group results to time tickets

When the group results are calculated, you can use the user exit "hyl_leistgrp2tls.hsc" to assign the group results to the day time tickets of the persons. The user exit is called for each time ticket with premium group. As import parameter, it includes the daily premium group result and as export parameter the time ticket data. For example, you can assign a performance efficiency rate to the time ticket that has been calculated for the group and you can enter a calculated standard time in the time ticket.

Import parameters:

Parameter	Type	Contents
PNR	N	Personnel number
PNR_*	*	Data of person, see section above.
ANR	C 40	Operation number of time ticket
ANR_*	*	Operation data, see section above.
MNR	C 20	Machine of time ticket
MNR_*	*	Data of time ticket machine, see section above
L_LART_*	*	Wage type data of time ticket, see section above.
LEISTGRP_*	*	Master data of premium group, see section above.
Data of premium group result		
VORG	N	Standard time in seconds
ZUSCHL	N	Bonuses in seconds. Is normally used for bonuses that refer to the standard time.
UPZ	N	Non-productive time in seconds
DAUER	N	Duration in seconds
AUSFZ	N	Downtime in seconds.
GKZ	N	Overhead cost times in seconds
GUTSCHR	N	Time credit in seconds. Is normally used for bonuses that refer to the actual time.
WARTEN	N	Waiting time in seconds
LEISTGRAD	F	Performance level in percent

PRKTO01	F	Premium account 1
...	...	
PRKTO20	F	Premium account 20
PRKZ01	C 20	Premium indicator 1
...	...	...
PRKZ05	C 20	Premium indicator 5

Export parameters:

Parameter	Type	Contents
L_	*	Data for time ticket, see section above.

1.5.6 Info function on the PZE terminal CT-WIN/CT-AIP

1.5.6.1 Overview

When the info is displayed on the PZE terminal, an information on the activities performed in the premium group is shown in addition to the account balances.

Flexitime	:	00:00
Flexitime	:	154:00
Leave account	:	27.00
05 B3P	102%	: 12:30
04 350	118%	: 7:30
04 B3P	122%	: 112:30

For more information on this info function, refer to the document describing the HR functions of the data collection software (status 2018: documents AIP-HRF and AIP-HRL).

You can change or disable the info display if you customize the incentive wage module using the user exit described in the following.

1.5.6.2 Formatting via user exit

1.5.6.3 Data rows

Use the user exit "hyl_info.hsc" to format the terminal info on your own, to show further data and to display total lines. This user exit is the equivalent to the user exit used to calculate the group results with personal group participation ("lpb00000.hsc", ART=P). This user exit only includes four other parameters that are used to format the info row.

Import parameters:

Parameter	Type	Contents
LEN_DIS_BEZ	N	Obsolete: some PZE terminals (DOS) send an information on the maximum length of the name. This information is provided here.
LEN_DIS_WERT	N	Obsolete: maximum length of value (normally 7)

Import/export parameters:

Parameter	Type	Contents
DIS_BEZ	max C100	Formatted name that is displayed on the terminal. If the name is empty, the row is not displayed on the terminal.
DIS_WERT	max C7	Formatted time value that is displayed on the terminal

1.5.6.4 Total line

All premium accounts (fixed and flexible) are totaled for the premium groups once a month. This sum total is then provided when the user exit "hyl_info.hsc" is called the next time. The user exit does not provide any information on the premium group and the persons. "SUM" is only assigned to the premium group LEISTGRP.

The name does not have a default formatting for the total line. Without formatting in the user exit, the total line is not shown.

See also the example below.

1.5.6.5 Example 1 – Extension of display

In the example that follows, the info displays the performance efficiency rate and additionally the premium account 1. And a total standard time is provided for the total line of premium account 10.

In the total line, the duration recorded and the standard time recorded is used to calculate a total performance efficiency rate that is shown in the total line on the terminal.

```
hydra basic;

// import data general
import ART          char(1);      // request type of script.
                                // P: script is executed for the calculation of the
                                // personal group participation.
import DAT          date;         // date
import PDAUER       long;         // working time of a person in seconds.
import LEISTGRP     char(10);     // premium group
```

```

export LEISTGRAD          double;      // performance level in percent
export PRKTO01            double;      // premium account 1 (PKZ)
export PRKTO10            double;      // premium account 10 (standard time)

// Import maximum length of name and value
import LEN_DIS_BEZ        long;        // maximum length of name
import LEN_DIS_WERT        long;        // maximum length of value (normally 7)

// Import/Export name and value
export DIS_BEZ             char(80);    // name of info
export DIS_WERT            char(8);     // value of info

/*-----*/
/* Main function */
/*-----*/
long main()
{
    code long;

    if( ( month( DAT ) < month( today() ) ) and
        ( day( today() ) > 7 ) ) )
    {
        DIS_BEZ = "";
        dprint( "nach dem siebten den Vormonat nicht mehr anzeigen." );
    }
    else
    {
        if( LEISTGRP <> "SUM" )
        {
            DIS_BEZ = (month( DAT ) using "&&") || " " ||
                (LEISTGRP stripped) || ": " ||
                (LEISTGRAD using "##&") || "%(" ||
                (PRKTO01 using "##&") || ")";
            PRKTO10 = PDAUER * LEISTGRAD / 100.0;
        }
        else
        {
            if( PDAUER > 0 )
            {
                LEISTGRAD = PRKTO10 / PDAUER * 100;
                DIS_BEZ = (month( DAT ) using "&&") || " " ||
                    ("Summe") || ": " ||
                    (LEISTGRAD using "##&") || "%";
            }
        }
    }

    return code;
}

/*-----*/

```

Output:

Flextime	:	00:00
Flexitime	:	154:00
Leave account	:	27.00
05 B3P: 57%(55)	:	12:30
05 total: 57%	:	12:30
04 350: 118%(95)	:	7:30
04 B3P: 122%(116)	:	112:30
04 total: 122%	:	120:00

Other example of a terminal display on CTWIN:

Information [0466]		
Bezeichnung		Wert
Gleitzeit		0:00
Flexzeit		159:45
Urlaubskonto		30.00
09 86002	122.53%	10:45
09 86003	112.83%	20:05
09 Summe	116.10%	30:50
F1 Seite↑ F2 Seite↓ F9 Stempelsätze laden  OK		

1.5.6.6 Example 2 – Suppressing display

```

hydra basic;

// // import data general
// import ART          char(1);      // request type of script.
//                               // P: script is executed for the calculation of the
//                               // personal group participation.
// import DAT          date;          // date
// import PDAUER        long;         // working time of a person in seconds.
//
// import LEISTGRP      char(10);     // premium group
//
// export LEISTGRAD      double;       // performance level in percent
// export PRKT001        double;       // premium account 1 (PKZ)
// ...
// export PRKT010        double;       // premium account 10 (standard time)
//
// // Import maximum length of name and value
// import LEN_DIS_BEZ    long;         // maximum length of name
// import LEN_DIS_WERT    long;         // maximum length of value (normally 7)
//
// Import/Export name and value
export DIS_BEZ          char(80);     // name of info
export DIS_WERT         char(8);      // value of info

/*-----*/
/* Main function */
/*-----*/
long main()
{
    // set name to empty --> output of the row is suppressed.
    DIS_BEZ = "";

    return 0;
}

/*-----*/

```

1.6 Calculating period results



This function is available as of MW 3.0, service pack 6 (end 2014). Older versions do not meet the software requirements. The user exit described below is not called.



With HYDRA systems MW 3.0 and an initial installation before February 2015, you must check if the HYDRA database fulfills the requirements before you use the function. If required, you must first execute the patches for MW 3.0 of service pack 6 status.

You can persist the results of complete settlement periods in the database for a further customer-specific processing for persons and premium groups. In HYDRA, the settlement periods of the Incentive Wage are fixed to calendar months.

As part of the wage calculation, the settlement period is identified that must be recalculated for the persons or premium groups. A settlement period must be recalculated if at least one day result of the settlement period has been recalculated.

A user exit is then called for each settlement period and person or premium group. With each call, the user exit can calculate one period result. To calculate the result, the user exit uses SQL database accesses that usually total the relevant day results. Then, the totaled result is calculated for the settlement period and assigned to the export variable. The wage calculation automatically saves the result. The HYDRA system provides examples of user exits for persons or premium groups.

1.6.1 Settlement period results for persons

User exit: hyl_pnrperiod_calc.hsc

Function: long final_calc()

Parameter of function: -none-

Return value: is not processed. The function must return the value 0.

Import parameters:

Parameter	Type	Contents
PNR	N	Personnel number
PERSON_*	*	Data of person, see section above.

Export parameter:

Parameter	Type	Contents
PERSONPERIOD_YEAR	N	Year of settlement period

PERSONPERIOD_PERIOD	N	Settlement period (calendar month)
PERSONPERIOD_ACC_DATB	D	First day of settlement period (first day of calendar month)
PERSONPERIOD_ACC_DATE	D	Last day of settlement period (last day of calendar month)
PERSONPERIOD_ACTUALTIME	F	(Actual) duration [h]
PERSONPERIOD_STANDARDTIME	F	Standard time [h]
PERSONPERIOD_PERFEFFRATE	F	Performance level
PERSONPERIOD_MINPERFLEVEL	F	Minimum day performance level of settlement period
PERSONPERIOD_MAXPERFLEVEL	F	Maximum day performance level of settlement period
PERSONPERIOD_DURATION_ADE	F	Duration ADE [h] (optional)
PERSONPERIOD_DURATION_PZE	F	Duration PZE [h] (optional)
PERSONPERIOD_DURATION_LLE	F	Duration LLE [h] (optional)
PERSONPERIOD_PRACC_01 to PERSONPERIOD_PRACC_30	F	Premium accounts that the user can define
PERSONPERIOD_PRATTR_01 to PERSONPERIOD_PRATTR_05	C 20	Premium indicators that the user can define

1.6.2 Settlement period results for premium groups

User exit: hyl_pgrppperiod_calc.hsc

Function: long final_calc()

Parameter of function: -none-

Return value: is not processed. The function must return the value 0.

Import parameters:

Parameter	Type	Contents
PERSON_*	*	Master data of premium group, see section above.

Export parameter:

Parameter	Type	Contents
PRGRPPERIOD_PRGRP	C 8	Premium group •
PRGRPPERIOD_YEAR	N	Year of settlement period
PRGRPPERIOD_PERIOD	N	Settlement period (calendar month)
PRGRPPERIOD_ACC_DATB	date	First day of settlement period (first day of calendar month)
PRGRPPERIOD_ACC_DATE	date	Last day of settlement period (last day of calendar month)
PRGRPPERIOD_ACTUALTIME	F	(Actual) duration [h]
PRGRPPERIOD_STANDARDTIME	F	Standard time [h]

PRGRPPERIOD_PERFEFFRATE	F	Performance level
PRGRPPERIOD_MINPERFLEVEL	F	Minimum day performance level of settlement period
PRGRPPERIOD_MAXPERFLEVEL	F	Maximum day performance level of settlement period
PRGRPPERIOD_OFFTIME	F	Downtime [h]
PRGRPPERIOD_PREMIUMAVERAGE	F	Time of premium average (overhead times) [h]
PRGRPPERIOD_CREDITNOTE	F	Time credit [h]. Is normally used for bonuses that refer to the actual time.
PRGRPPERIOD_UNPRODTIME	F	Non-productive time [h]
PRGRPPERIOD_WAITINGTIME	F	Waiting time [h]
PRGRPPERIOD_BONUSES	F	Bonuses [h]. Is normally used for bonuses that refer to the standard time.
PERSONPERIOD_PRACC_01 to PERSONPERIOD_PRACC_30	F	Premium accounts that the user can define
PERSONPERIOD_PRATTR_01 to PERSONPERIOD_PRATTR_05	C 20	Premium indicators that the user can define

1.7 Interface to payroll accounting

1.7.1 Overview

The interface to payroll accounting is an interface that can be used universally to transfer all relevant data from the HYDRA Incentive Wage and HYDRA Time and Attendance to any payroll accounting system. The payroll accounting system only requires a defined possibility to read a sequential ASCII file.

The customer or the MPDV Consulting can specify the format of the output file via user exit. Via user exit, the customer or the MPDV Consulting can also specify the contents of the output file that use the available data. For example, the degree of aggregation can be specified (per day or month, for each cost center or none). It is also possible to make additional calculations, for example average values.

1.7.2 Data that can be processed

1.7.2.1 Incentive Wage - individual time tickets

This data is equivalent to the data that is shown on the HYDRA console in the LLE menu, item Reports/time ticket log.

This data is the basis for the payment of persons with an individual allocation of piecework or time wage.

1.7.2.2 Incentive Wage - personal group participation

This data is shown on the HYDRA console in the LLE menu, item Reports/Group reports/Personal group participation.

This data is the basis for the payment of persons with group-related allocation.

1.7.2.3 Incentive Wage - monthly group results

This data is shown on the HYDRA console in the LLE menu, item Reports/Group reports/Monthly group results.

This data does not refer to a person and is only required in the interface if the payroll accounting system uses the group results identified in HYDRA to calculate other values or if the results are used for calculation purposes.

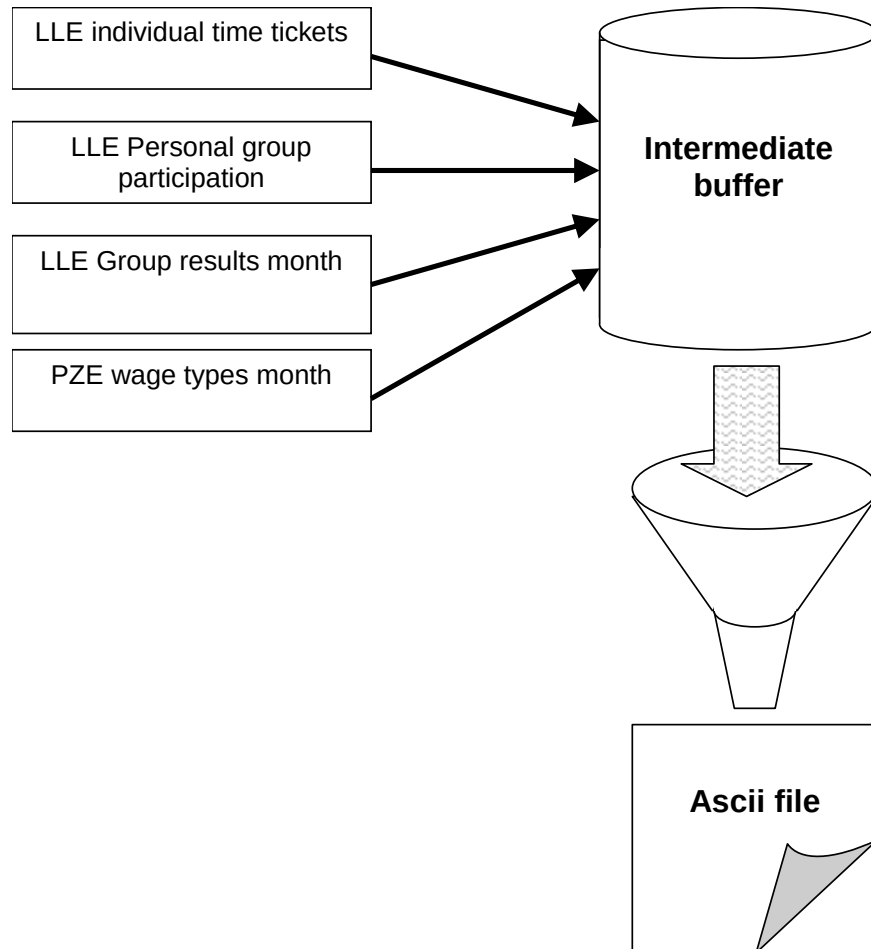
1.7.2.4 Time & Attendance - wage types

This data is shown on the HYDRA console in the PZE menu, item Reports/Monthly wage types. If this HYDRA Time and Attendance data is used for calculation, this data is the basis for the payment of persons.

1.7.3 Procedure to create interface file

All relevant data is transferred one after the other to a user exit. In this user exit, the required data types are selected. From this data, the key values (e.g. personnel number and wage type) and data (e.g. duration or monetary value) are identified and written in a buffer. The write process is explicitly triggered in the user exit and using one single input data record, any number of entries can be created in the buffer (it might also be useful to create no entry at all).

In the next step, the data of the buffer is aggregated to key values and the data fields are totaled. This aggregated data is transferred to a user exit. The user exit converts the data into a character string that is written in the interface file and triggers this write process in the interface file.



1.7.4 User exits

1.7.4.1 Initialization

[Initialization available as of hyl_rck72.exe|out 7.2.1.10]

User exit "lrck1000.hsc", function `init()`.

At start of processing, the function **`init()`** is called in user exit `lpv1000.hsc`. In this function, you use the export variables **`USES_ART`** to control which data types are processed in the interface. Only the data required to create the interface file must be requested.

The import and export variables are described in the section below.

With the function **`init()`** you must make sure that only the import variables `DAT_VON` and `DAT_BIS` are filled with values. The other import variables are empty or 0.

```
// -----
// Function init (as of 08/2010)
//
// initial control of processing. The function is called once before
// start of data processing. At this time,
// only the import variables DAT_VON and DAT_BIS are filled with useful values.
//
// -----
long init()
{
    // The export variable USES_ART controls, which data is provided to the interface.
    USES_ART = "ELS GRP GRE PZE";

    return 0;
}
```

1.7.4.2 Step 1: Data collection

User exit "lrck1000.hsc", function main().

This user exit controls the data collection

- 1) which data records are relevant for the interface
- 2) which fields of the data are relevant key fields for the interface
- 3) which fields of the data are relevant data fields that are totaled for the interface.

The parameter ART specifies the data type for which the user exit is called to process the data.

Note: the import parameters of the user exit are filled with reference to the parameter ART. For example, in the processing of Time and Attendance data, no data of a premium group is available.

Parameter	Type	Contents
VERARBKZ	C10	Field <i>Processing</i> of the selection criteria when you create the interface file.
ART	C 3	ELS: Individual time tickets (from individual allocation, AKK, ZL, GK, ...) GRP: Personal group participation (with relevant group results) GRE: Group results (without reference to a person) PZE: Wage type postings of the Time and Attendance PZM: PZE monthly results. Available as of 09/2010. PNR: HR master data. Available as of 04/2012 (8.1.1.23). Only import variables of the HR master data are populated.
DAT_VON	D	Start date of evaluation period
DAT_BIS	D	End date of evaluation period
SATZNR	N	Unique sequence number of data record, starting with 1

Export parameter

Parameter	Type	Contents
USES_ART	C 50	Only relevant if the interface is initialized via function init().

		You can enter the abbreviations described for parameter ART in this variable. Separate by space character. This variable then controls which data is selected by the interface program. For example: USES_ART = "ELS GRP";
--	--	---

Further import parameters

Data of the person at the end of the evaluation period (only with ART = ELS, GRP, PZE and PNR)		
PNR	N	Person: personnel number
PNR_NACHNAME	C 40	Person: last name. (as of version 8.1.1.23 04/2012)
PNR_VORNAME	C 20	Person: first name. (as of version 8.1.1.23 04/2012)
PNR_NAME	C 62	Person: last name, first name. (as of version 8.1.1.23 04/2012)
PNR_PRKZ	C 1	Person: premium indicator
PNR_ABT	C 8	Person: department
PNR_BER	C 8	Person: area
PNR_EINTRITT	D	Person: date of joining
PNR_FIR	C 4	Person: company
PNR_GEBDAT	D	Person: date of birth
PNR_GESCHLECHT	C 1	Person: gender M/W (male/female)
PNR_INFODAT_1	D	Person: free date field 1
...	...	...
PNR_INFODAT_5	D	Person: free date field 5
PNR_INFOTXT_01	C 40	Person: free text field 1
...	...	...
PNR_INFOTXT_20	C 40	Person: free text field 20
PNR_INFOWERT_1	N	Person: free number field 1
...	...	...
PNR_INFOWERT_5	N	Person: free number field 5
PNR_KST	C10	Person: cost center
PNR_LEISTGRP	C 8	Person: regular premium group of person
PNR_ANTFAKTLBON	N	Person: proport. factor for incentive bonus

Time ticket of person (only with type ELS, GRP and PZE, special assignment with type PZM, see below)		
L_DAT	D	Date
L_ANR	C 40	Order number. (not populated with ART PZE and GRP).

L_SKNR	N	Shift number (not populated with ART PZE and GRP).
L_MNR	C 20	Machine number
L_BMKNR	N	Number of resource performance account (only populated with ART ELS)
L_ZEIART	C 3	Time type of time ticket. (not populated with ART PZE).
L_LART	C 4	Wage type of time ticket (not populated with ART GRP).
L_TE	F	Target t_e of the time ticket in seconds for 1000 pieces (not populated with ART PZE and GRP).
L_TR	F	Target t_r of the time ticket in seconds for 1000 pieces (not populated with ART PZE and GRP).
L_GUT	F	Yield of time ticket (not populated with ART PZE and GRE).
L_AUS	F	Scrap of time ticket (not populated with ART PZE and GRE).
L_DAUER	N	Type ELS: duration of time ticket in seconds [s] Type GRP: personal group participation with person-related time proportions [s] Type PZE: Total of attendance time and absence time [s]
L_VGZ	N	Standard time of time ticket in seconds (not populated with ART PZE and GRE).
L_ZUSCHL	N	Bonus time of time ticket in seconds (not populated with ART PZE and GRE).
L_ZUSCHLGR	N	Bonus reason of time ticket (not populated with ART GRP).
L_KST	C10	Cost center of time ticket (not populated with ART GRP).
L_LOHNGRP	C 4	Wage group of time ticket (not populated with ART GRP).
L_ZEITGRAD	F	Performance efficiency rate of time ticket in percent. Performance efficiency rates are only calculated for time tickets of the AKK time type (piecework). (not populated with ART PZE).
L_REFNR	C 20	Reference number of time ticket. Is usually empty, can be shown for customer-specific information in the time ticket log. (not populated with ART PZE and GRP).
L_PRKTO_01	F	Premium account of time ticket 1 that can be defined by user. With type PZE, this parameter contains the attendance time of the month recorded for the wage type of an employee. (not populated with ART GRP).
L_PRKTO_02	F	Premium account of time ticket 2 that can be defined by user. With type PZE, this parameter contains the absence time of the month recorded for the wage type of an employee. (not populated with ART GRP).
L_PRKTO_03	F	Premium account of time ticket 3 that can be defined by user. With type PZE, this parameter contains the number of full absence days of the month recorded for the payment day type with the wage type number of an employee. (not populated with ART GRP).
L_PRKTO_04	F	Premium account of time ticket 4 that can be defined by user. With type PZE, this parameter contains the number of partial absence days of the month recorded for the payment day type

		with the wage type number of an employee. (not populated with ART GRP).
L_PRKTO_05	F	Premium account of time ticket 5 that can be defined by user. With type PZE, this parameter contains the number of leave days taken (multiplied by factor 10) in the month. (not populated with ART GRP).
L_PRKTO_06	F	Premium account of time ticket 6 that can be defined by user. With type PZE, this parameter contains the full attendance days of an employee in the month (independent of wage type). (not populated with ART GRP).
L_PRKTO_07	F	Premium account of time ticket 7 that can be defined by user. With type PZE, this parameter contains the full attendance hours of an employee in the month (independent of wage type). (not populated with ART GRP).
...	...	...
L_PRKTO_10	F	Premium account of time ticket 10 that can be defined by user (not populated with ART PZE and GRP)
L_PRKZ_01	C 20	Premium account of time ticket 1 that can be defined by user (not populated with ART PZE and GRP)
...	...	...
L_PRKZ_05	C 20	Premium account of time ticket 5 that can be defined by user (not populated with ART PZE and GRP)

Time ticket of person with type PZM

(Only the parameters populated are listed. Other parameters are not populated and are empty or 0).

L_DAT	D	End of PZE settlement period
L_LART	C 4	Wage type
L_KST	C10	Cost center
L_DAUER	N	Total time of person in month
L_VGZ	N	Target time in hours in the month
L_PRKTO_01	F	Attendance time of person in month
L_PRKTO_02	F	Absence
L_PRKTO_03	F	Days present
L_PRKTO_04	F	Number of days with absences
L_PRKTO_05	F	Target time in hours in the month
L_PRKTO_06	F	Number of days with target time

Wage type data

L_LART_*	*	Master data of wage type, refer to section further ahead. The master data of the wage types is available for all data records that contain a wage type.
----------	---	---

LEISTGRP_LART_	*	Data of the wage type of a premium group (reserved)
Data of the premium group (only with ART = GRE and GRP, also ART = ELS with group time tickets. With ART = PNR populated with premium group of HR master data.)		
LEISTGRP	C10	Premium group •
LEISTGRP_*	*	Data of premium group, see section above.
Results of the premium groups (only with ART = GRE and GRP, also ART = ELS with group time tickets. Also populated with ART = PNR with the group results in the evaluation period recorded for the premium group of the HR master data at the end of evaluation period).		
VORG	F	Standard time in seconds
ZUSCHL	F	Bonuses in seconds. Is normally used for bonuses that refer to the standard time.
UPZ	F	Non-productive time in seconds
DAUER	F	Duration in seconds
AUSFZ	F	Downtime in seconds.
GKZ	F	Overhead cost times in seconds
GUTSCHR	F	Time credit in seconds. Is normally used for bonuses that refer to the actual time.
WARTEN	F	Waiting time in seconds
LEISTGRAD	F	Performance level in percent
PRKTO01	F	Premium account 1
...	...	
PRKTO30	F	Premium account 30
PRKZ01	C 20	Premium indicator 1
...	...	...
PRKZ05	C 20	Premium indicator 5

Callback function:

This user exit includes a callback function INSERT_DATA. This callback function expects a so-called BAPI string as second parameter that contains the contents of the buffer. The following example shows the use of the callback function and shows the possible fields for the buffer.

```

data = add_bapi_val( "", "DAT", DAT_VON );
data = add_bapi_val( data, "JAHR", jahr );
data = add_bapi_val( data, "PER", periode );
data = add_bapi_val( data, "FIR", PNR_FIR );
data = add_bapi_val( data, "PNR", PNR );
data = add_bapi_val( data, "LART", L_LART );
data = add_bapi_val( data, "KST", PNR_KST );
data = add_bapi_val( data, "KEYTEXT:1", ART );
...
data = add_bapi_val( data, "KEYTEXT:10", LEISTGRP );
data = add_bapi_val( data, "KEYWERT:1", xxx );

```

```

...
data = add_bapi_val( data, "KEYWERT:5", xxx );
data = add_bapi_val( data, "DATAWERT:1", DAUER );
...
data = add_bapi_val( data, "DATAWERT:30", 0.0 );
data = add_bapi_val( data, "MINWERT:1", dat_long ); // as of 11/2005
...
data = add_bapi_val( data, "MINWERT:10", 0.0 );      // as of 11/2005
data = add_bapi_val( data, "MAXWERT:1", dat_long ); // as of 11/2005
...
data = add_bapi_val( data, "MAXWERT:10", 0.0 );      // as of 11/2005
code = CallBack("INSERT_DATA", data );

```

For information on the data types and the maximum field lengths, refer to the user exit description and the output of aggregated data below.

Key values are all fields **except** the fields "DATAWERT:xx", "MINWERT:xx" and "MAXWERT:xx".

The fields "DATAWERT:xx" are totaled in a second step after the key values, and of the fields "MINWERT:xx" and "MAXWERT:xx" the smallest or the greatest value is identified.

1.7.4.3 Step 2: Output of aggregated data

User exit "lrck2000.hsc".

This user exit is used to write the aggregated data of the buffer in the interface file in step 2 of the interface generation. To this end, further calculations can be made in the user exit.

The data is aggregated to key values and processed. The following sorting applies: company, personnel number, date, year, period, KEYTEXT_1 to 5, wage type, cost center, KEYTEXT_6 to 10, KEYWERT_1 to 5.

The import parameters are the fields of the buffer that have been populated by user exit lrck1000.hsc in step 1.

Parameter	Type	Contents
VERARBKZ	C10	Field <i>Processing</i> of the selection criteria when you create the interface file.
DAT_VON	D	Start date of evaluation period
DAT_BIS	D	End date of evaluation period
SATZNR	N	Unique sequence number of data record
PNR	N	Intended for personnel number
FIRMA	C 4	Intended for company
NACHNAME	C 40	Contains the last name, if PNR contains a valid personnel number.

VORNAME	C 20	Contains the first name, if PNR contains a valid personnel number.
NAME	C 62	Contains the first and last name separated by comma, if PNR contains a valid personnel number.
DATUM	D	Intended for a date.
JAHR	N	Intended for the settlement year
PERIODE	N	Intended for the settlement period (month)
LART	C 4	Intended for wage type
LART_*	*	Master data of wage type, refer to section further ahead.
KST	C10	Intended for cost center
KEYTEXT_1 to 10	C 20	Character string for further keys that can be used by the user
KEYWERT_1 to 5	F	Numeric values for further keys that can be used by the user
DATAWERT_1 to 30	F	Free data values that can be used. These values are totaled using the single values of all key fields mentioned above.
MINWERT_1 to 10	F	Free data values that can be used. The smallest values of these values are identified using the single values of all key fields mentioned above. Available from 11/2005.
MAXWERT_1 to 10	F	Free data values that can be used. The largest values of these values are identified using the single values of all key fields mentioned above. Available from 11/2005.

The user exit does not contain any **export parameters**.

Callback function:

This user exit includes several callback functions. All callback functions expect a character string as second parameter.

OUTPUT Outputs the character string in the interface file and shows it on the HYDRA console in the dialog to create the interface file.

OUTPUT.DATA Outputs the character string only in the interface file. An output on the HYDRA console is not performed.

OUTPUT.DISPLAY Outputs the character string only on the HYDRA console in the dialog to create the interface file. An output in the interface file is not performed.

1.8 User field configuration with premium accounts

1.8.1 User field key

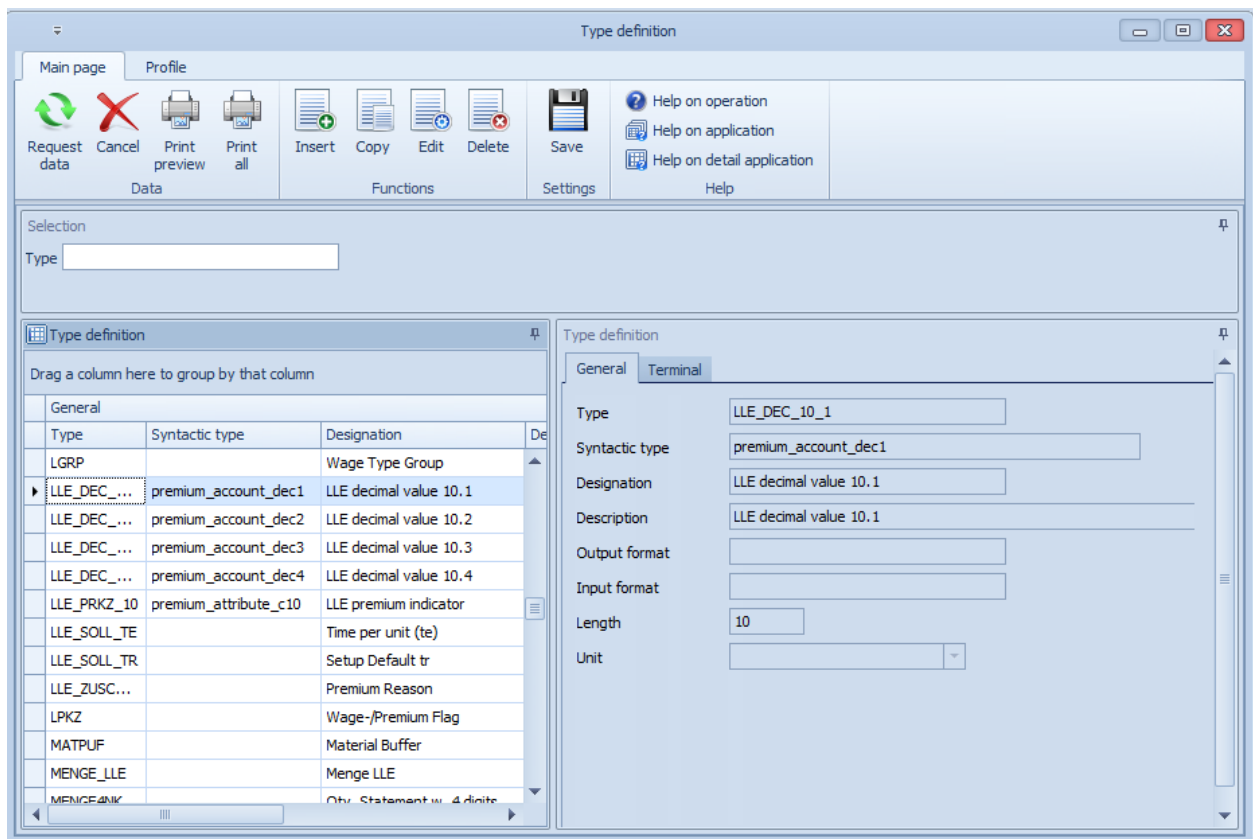
The incentive wage based on formulas requires the following user field keys. The user field keys are fixed, other user field keys are not processed.

Object	User field key
LEISTGRP	SYSTEM
LEISTGRPTG	SYSTEM
LLEPNRTAG	SYSTEM
TLS	SYSTEM

1.8.2 Type definitions

Default type definitions are specified. They must not be changed. If you want to use deviating type definitions, you must create customer-specific type definitions.

The following other type definitions are provided for the LLE:



The screenshot shows the 'Type definition' window in SAP. The window has a menu bar with 'Main page' and 'Profile'. Below the menu bar is a toolbar with icons for 'Request data', 'Cancel', 'Print preview', 'Print all', 'Insert', 'Copy', 'Edit', 'Delete', 'Save', and 'Help'. The main area is divided into two panes. The left pane shows a list of type definitions with columns 'Type', 'Syntactic type', and 'Designation'. The right pane shows the details of the selected type definition, 'LLE_DEC_10_1', with fields for 'Type', 'Syntactic type', 'Designation', 'Description', 'Output format', 'Input format', 'Length', and 'Unit'.

Type	Syntactic type	Designation
LGRP		Wage Type Group
LLE_DEC_10_1	premium_account_dec1	LLE decimal value 10.1
LLE_DEC_10_2	premium_account_dec2	LLE decimal value 10.2
LLE_DEC_10_3	premium_account_dec3	LLE decimal value 10.3
LLE_DEC_10_4	premium_account_dec4	LLE decimal value 10.4
LLE_PRKZ_10	premium_attribute_c10	LLE premium indicator
LLE_SOLL_TE		Time per unit (te)
LLE_SOLL_TR		Setup Default tr
LLE_ZUSC...		Premium Reason
LPKZ		Wage-/Premium Flag
MATPUF		Material Buffer
MENGE_LLE		Menge LLE
MENGE_LLE		Qty. Statement w. 4 digits

Details of 'LLE_DEC_10_1':

Type	LLE_DEC_10_1
Syntactic type	premium_account_dec1
Designation	LLE decimal value 10.1
Description	LLE decimal value 10.1
Output format	
Input format	
Length	10
Unit	

Older systems might use deviating type definitions: LEISTGRPFAKT??, LEISTGRPMOD?, LEISTGRPTGPRKTO??, LEISTGRPTGPRKZ??, TLSPRKTO?? and TLSPRKZ?. These type definitions should not be used anymore.

1.8.3 User fields

In the user fields in field *Designation*, you can select the label text of the user fields on the MOC.

Object type	User field key	Field ID	Default field type	Purpose
LEISTGRP	SYSTEM	1 to 30	LLE_DEC_10_3	Numeric default values of premium groups
LEISTGRP	SYSTEM	101 to 105	LLE_PRKZ_10	Alphanumeric default values of premium groups
LEISTGRPTG	SYSTEM	1 to 30	LLE_DEC_10_3	Premium accounts in the result of premium groups
LEISTGRPTG	SYSTEM	101 to 105	LLE_PRKZ_10	Premium indicator in the result of premium groups
LLEPNRTAG	SYSTEM	1 to 30	LLE_DEC_10_3	Premium accounts in the person day performance
LLEPNRTAG	SYSTEM	101 to 105	LLE_PRKZ_10	Premium indicator in the person day performance
TLS	SYSTEM	1 to 30	LLE_DEC_10_3	Premium accounts in the time tickets of the person
TLS	SYSTEM	101 to 105	LLE_PRKZ_10	Premium indicators in the time tickets of the person.